

PROJECT-3:INPUT

```
GNU nano 8.7.1 practical03.c *

printf("\n[Stage 3: Child After Sleep]\n");
printf("Child PID : %d\n", getpid());
printf("Child PPID : %d\n", getppid());
printf("State : Running\n");

printf("\nChild is terminating...\n");

exit(0);
}

else
{
    // Parent process
    printf("\n[Stage 2: Parent Process]\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("Child PID : %d\n", pid);
    printf("State : Running\n");

    printf("\nParent is waiting for child to finish...\n");
    printf("Parent State: Waiting\n");

    wait(&status);

    printf("\n[Stage 4: Parent After Child Termination]\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("State : Running\n");

    printf("\nChild has terminated successfully.\n");
    printf("Parent is now terminating.\n");
}

return 0;
}
```

^G Help **^O** Write Out **^F** Where Is **^K** Cut **^T** Execute **^C** Location
^X Exit **^R** Read File **^_** Replace **^U** Paste **^J** Justify **^/_** Go To Line

```
GNU nano 8.7.1 practical03.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("=== Process State Demonstration ===\n");

    printf("\n[Stage 1: Parent Before fork()]\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    printf("State : Running\n");

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    else if (pid == 0)
    {
        // Child process
        printf("\n[Stage 2: Child Process]\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());
        printf("State : Running\n");

        printf("\nChild is now going to sleep...\n");
        printf("State : Waiting/Sleeping\n");
    }
}
```

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OUTPUT

```
administrator@VBCO-LAP-29: ~ X + v
administrator@VBCO-LAP-29:~/OSLab/fork$ gcc practical03.c -o practical03
administrator@VBCO-LAP-29:~/OSLab/fork$ ./practical03
=== Process State Demonstration ===

[Stage 1: Parent Before fork()]
Parent PID   : 1579
Parent PPID  : 1428
State        : Running

[Stage 2: Parent Process]
Parent PID   : 1579
Parent PPID  : 1428
Child PID    : 1580
State        : Running

Parent is waiting for child to finish...
Parent State: Waiting

[Stage 2: Child Process]
Child PID    : 1580
Child PPID   : 1579
State        : Running

Child is now going to sleep...
State        : Waiting/Sleeping

[Stage 3: Child After Sleep]
Child PID    : 1580
Child PPID   : 1579
State        : Running
Child is terminating...

[Stage 4: Parent After Child Termination]
Parent PID   : 1579
Parent PPID  : 1428
State        : Running
Child has terminated successfully.
Parent is now terminating.
```